PROBLEM SET 4

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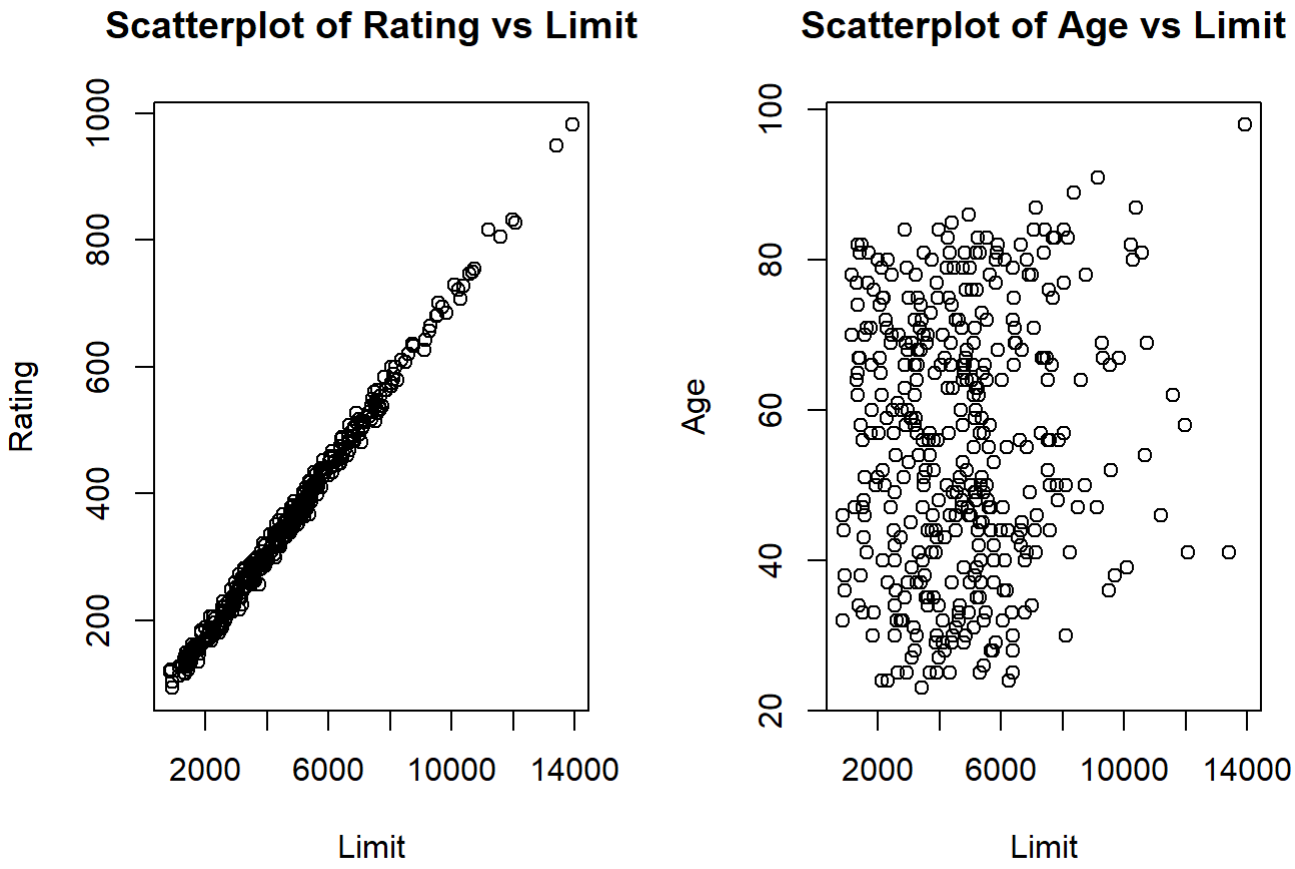
Problem 1: Problem to demonstrate multicollinearity

```
rm(list=ls())
library(ISLR)
## Warning: package 'ISLR' was built under R version 4.3.3
attach(Credit)
head(Credit)
##   ID  Income Limit Rating Cards Age Education Gender Student Married Ethnicity
## 1  1  14.891  3606    283     2  34        11   Male      No     Yes Caucasian
## 2  2 106.025  6645    483     3  82        15 Female     Yes     Yes     Asian
## 3  3 104.593  7075    514     4  71        11   Male      No      No     Asian
## 4  4 148.924  9504    681     3  36        11 Female      No      No     Asian
## 5  5  55.882  4897    357     2  68        16   Male      No     Yes Caucasian
## 6  6  80.180  8047    569     4  77        10   Male      No      No Caucasian
##   Balance
## 1     333
## 2     903
## 3     580
## 4     964
## 5     331
## 6    1151
df=Credit[,c(3,4,6,12)]
head(df)
##   Limit Rating Age Balance
## 1  3606    283  34     333
## 2  6645    483  82     903
## 3  7075    514  71     580
## 4  9504    681  36     964
## 5  4897    357  68     331
```

```
## 6  8047    569  77    1151
#(a)
par(mfrow=c(1,2))
plot(Limit,Rating,main="Scatterplot of Rating vs Limit")
plot(Limit,Age,main="Scatterplot of Age vs Limit")
```




```
par(mfrow=c(1,1))
```

Comment:

Age vs Limit:

1. Points are widely scattered.
2. Age does not appear strongly related to credit limit.

Rating vs Limit:

1. Strong positive linear relationship.
2. As Rating increases, Limit increases almost perfectly linearly.

```
#(b)
fit1=lm(Balance~Age+Limit)
fit2=lm(Balance~Rating+Age+Limit)
```

```
fit3=lm(Balance~Rating+Limit)

library(stargazer)

##

## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statisti
cs Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(fit1,fit2,fit3,type="text",out="f2.txt")

##

## ==============================================================================
==============

##                                                   Dependent variable:

##                      ---------------------------------------------------------
--------------

##                                                          Balance

##                                 (1)                       (2)
(3)

## ------------------------------------------------------------------------------
--------------

## Rating                                                   2.310**                  2.
202**

##                                                          (0.940)                  (0
.952)

##

## Age                          -2.291***                -2.346***

##                               (0.672)                  (0.669)

##

## Limit                         0.173***                  0.019                    0
.025

##                               (0.005)                  (0.063)                  (0
.064)

##

## Constant                    -173.411***              -259.518***              -377
.537***

##                               (43.828)                 (55.882)                 (4
5.254)

##

## ------------------------------------------------------------------------------
--------------

## Observations                    400                      400
400

## R2                            0.750                    0.754                    0
.746

## Adjusted R2                   0.749                    0.752                    0
.745
```

```
## Residual Std. Error    230.532 (df = 397)       229.080 (df = 396)       232.320
(df = 397)

## F Statistic         594.988*** (df = 2; 397) 403.718*** (df = 3; 396) 582.820***
(df = 2; 397)

## ================================================================================
==============

## Note:                                                           *p<0.1; **p<0
.05; ***p<0.01
```

Observed Marked Differences :

The major difference appears in Model (ii):

In Model (i), Limit is highly significant

In Model (iii), Rating is highly significant

In Model (ii)** (when both Rating and Limit are included) one of them becomes insignificant or changes sign

```
#(c) VIF

library(car)

## Loading required package: carData

vif(fit1)

##      Age    Limit

## 1.010283 1.010283

vif(fit2)

##     Rating        Age      Limit

## 160.668301   1.011385 160.592880

vif(fit3)

##   Rating    Limit

## 160.4933 160.4933
```

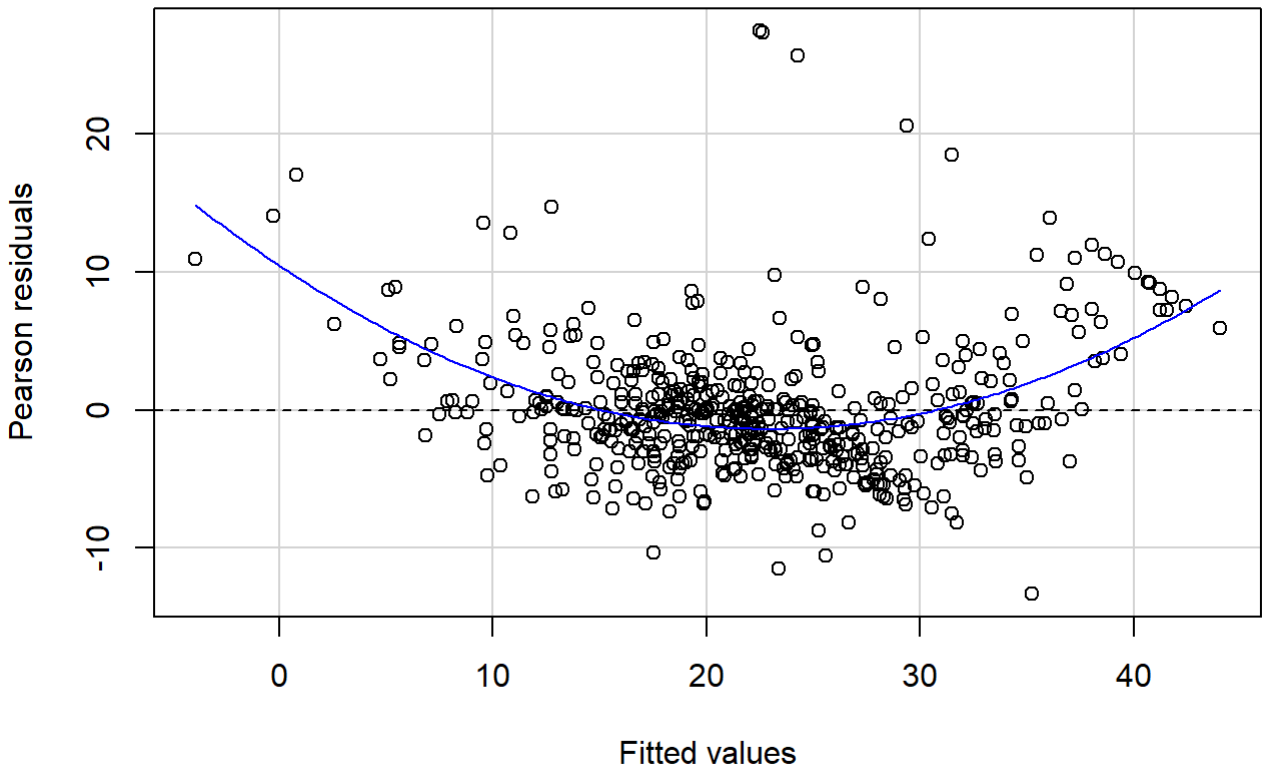
We observe: Very high VIF for Limit and Rating ,while low VIF for Age. Hence, strong multicollinearity exists between them.

Problem 2 :

```
rm(list=ls())

library(MASS)

head(Boston)

##      crim zn indus chas   nox    rm  age    dis rad tax ptratio  black lstat

## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3 396.90  4.98

## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8 396.90  9.14

## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8 392.83  4.03

## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7 394.63  2.94

## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7 396.90  5.33
```

```
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7 394.12  5.21
##   medv
## 1 24.0
## 2 21.6
## 3 34.7
## 4 33.4
## 5 36.2
## 6 28.7
dim(Boston)
## [1] 506  14
df=Boston
df.cont=Boston
df.cont$chas = NULL
df.cont$rad = NULL
model=lm(medv~.,data=df.cont)
summary(model)
##
## Call:
## lm(formula = medv ~ ., data = df.cont)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.3315  -2.8771  -0.6792   1.6858  27.4744
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.970e+01  5.051e+00   5.879 7.59e-09 ***
## crim        -7.010e-02  3.269e-02  -2.144 0.032482 *
## zn           3.989e-02  1.409e-02   2.831 0.004835 **
## indus       -4.198e-02  6.080e-02  -0.691 0.490195
## nox         -1.458e+01  3.899e+00  -3.740 0.000206 ***
## rm           4.188e+00  4.255e-01   9.843  < 2e-16 ***
## age         -1.868e-03  1.359e-02  -0.137 0.890696
## dis         -1.503e+00  2.059e-01  -7.301 1.15e-12 ***
## tax          8.334e-04  2.386e-03   0.349 0.727038
## ptratio     -8.738e-01  1.323e-01  -6.607 1.02e-10 ***
## black        8.843e-03  2.763e-03   3.200 0.001461 **
## lstat       -5.267e-01  5.224e-02 -10.083  < 2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.899 on 494 degrees of freedom
## Multiple R-squared:  0.7225, Adjusted R-squared:  0.7163
## F-statistic: 116.9 on 11 and 494 DF,  p-value: < 2.2e-16
residualPlot(model)
```




```
std.resid=rstandard(model)
```

We observe : A few observations have large studentized residuals → response outliers

Some observations have high leverage (extreme crim or lstat values) using hat- values.

A small number exceed Cook's distance threshold → influential points